

Computer Graphics (CSC214)

► Unit - 08: Illumination Models & Surface Rendering Techniques

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Credit Hours: 3hrs

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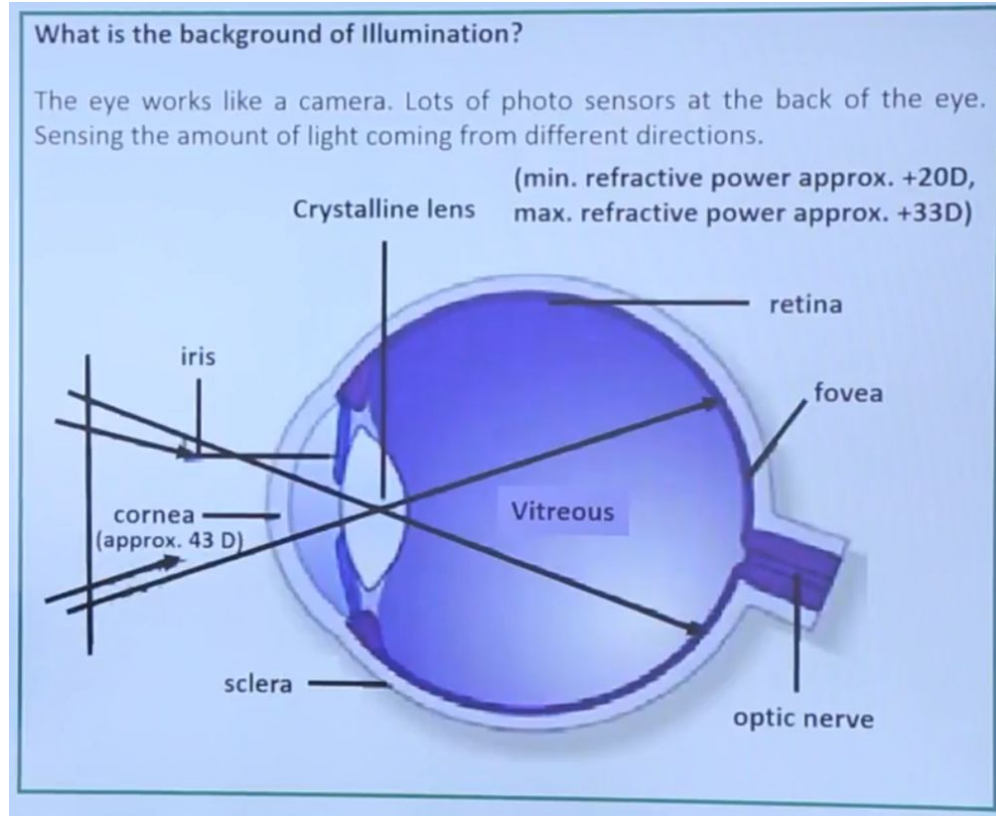
► Introduction

8.1

► Illumination Models:

- ❖ Illumination model is **used to calculate** the intensity of light **reflected from** a point on a surface.
- ❖ **Surface rendering** uses the intensity calculations **from the illumination model** to determine the light **intensity at all pixels** in the image, by possibly, considering light propagation between surfaces in the scene.

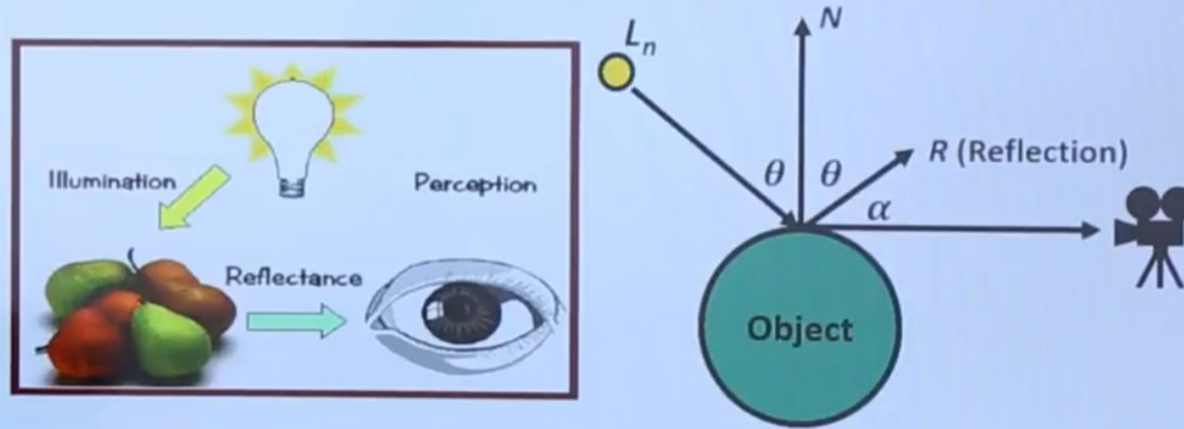
► Background of Illumination:



► Background of Illumination:

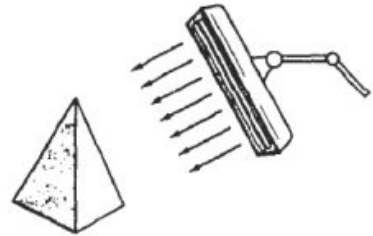
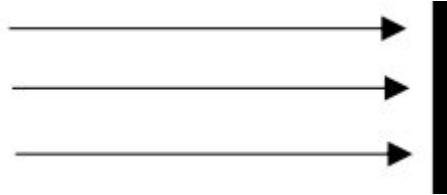
► What affects the light that comes into the eyes?

- The position of the point
- The position of the light
- Color and intensity of the light
- Camera vector
- Normal vector of the surface at the vertex
- Physical characteristics of the object (reflectance model, color etc.)



► Light Sources:

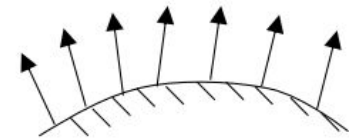
- ❖ Light sources are referred as **light emitting** object and **light reflectors**.
 - **Point source:** A point light source emits light equally in all directions from a single point.
 - **Parallel source:** Light rays are all parallel. May be modeled as a point source at infinity (the sun).
 - **Distributed source:** All light rays originate at a finite area in space. A nearby sources such as fluorescent light.



► Types of Reflection:

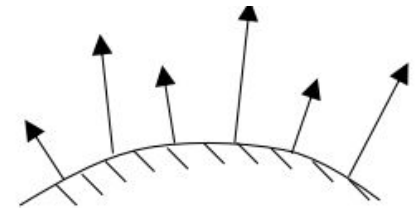
❖ Specular Reflections:

- Surfaces appear **equally bright from all viewing angles** since they reflect light with **equal intensity in all directions**. E.g. shiny surfaces
- Position of viewer is **not important**.
- Specular reflection often causes intense, bright glare, such as on a calm, sunny day on water.



❖ Diffuse Reflection:

- Light reflected with **unequal intensity**. E.g. rough surface.
- Position of viewer is important.
- Diffuse reflection allows for even, comfortable viewing of objects, such as reading a book.

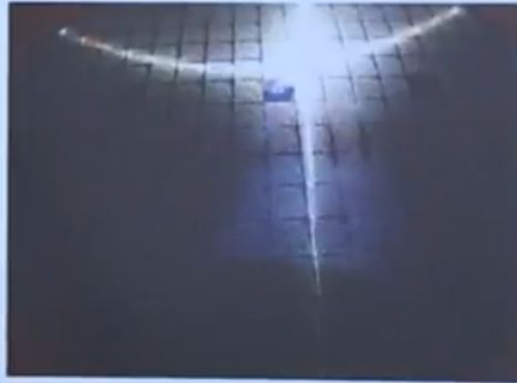
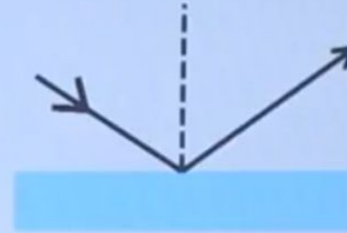


► Specular Reflection:

- ✓ Direct reflections of light source off shiny object.

- ✓ The object has a very smooth surface.

specular highlight on object



► Types of Reflection:

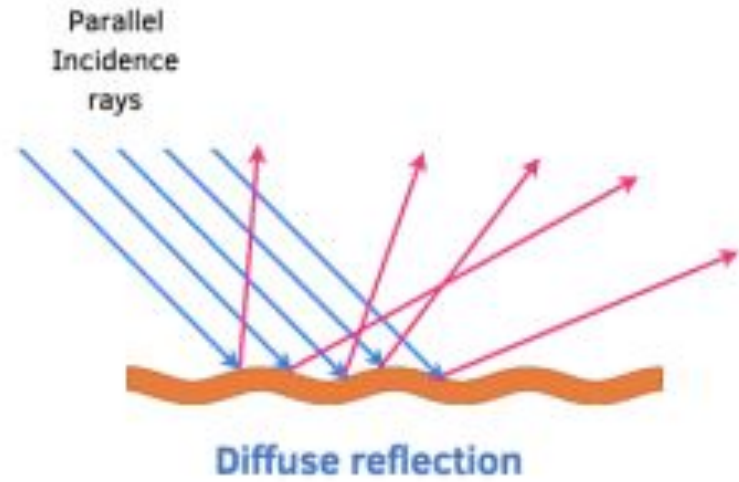
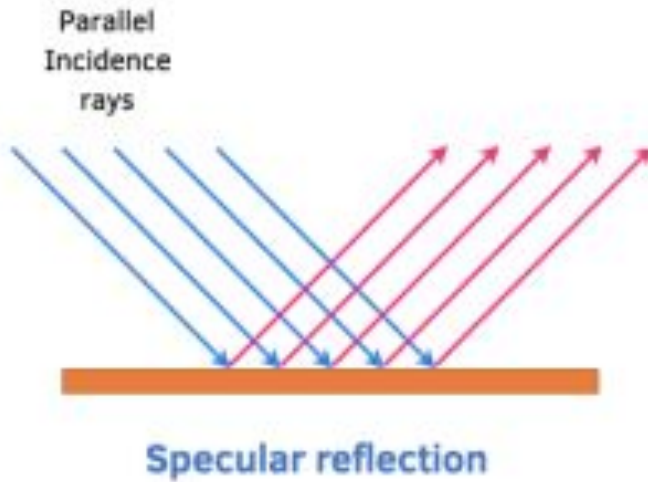


Fig 5. Types of Reflection: Specular and Diffuse Reflection. When Specular reflection occurs each ray has the **same** angle of incidence and reflection. When diffuse reflection occurs each ray has its **own** angle of incidence and reflection.

► Illumination model / Lighting model / Shading model :

- ❖ An illumination model **is a formula in variables** associated to the surface properties and light conditions **to calculate the intensity of light reflected from a point** on a surface.
- ❖ Based on standard lighting conditions in a scene, some **illumination models are**:
 - Ambient Light
 - Diffuse Reflection
 - Specular reflection and Phong model

► Illumination Model ► **Ambient light** :

Ambient light:

Surface getting light from various reflected source i.e. light not coming directly from a light source but coming after getting reflected from other surface.

Ambient light has no spatial or directional characteristics and amount on each object is a constant for all surfaces and all directions.

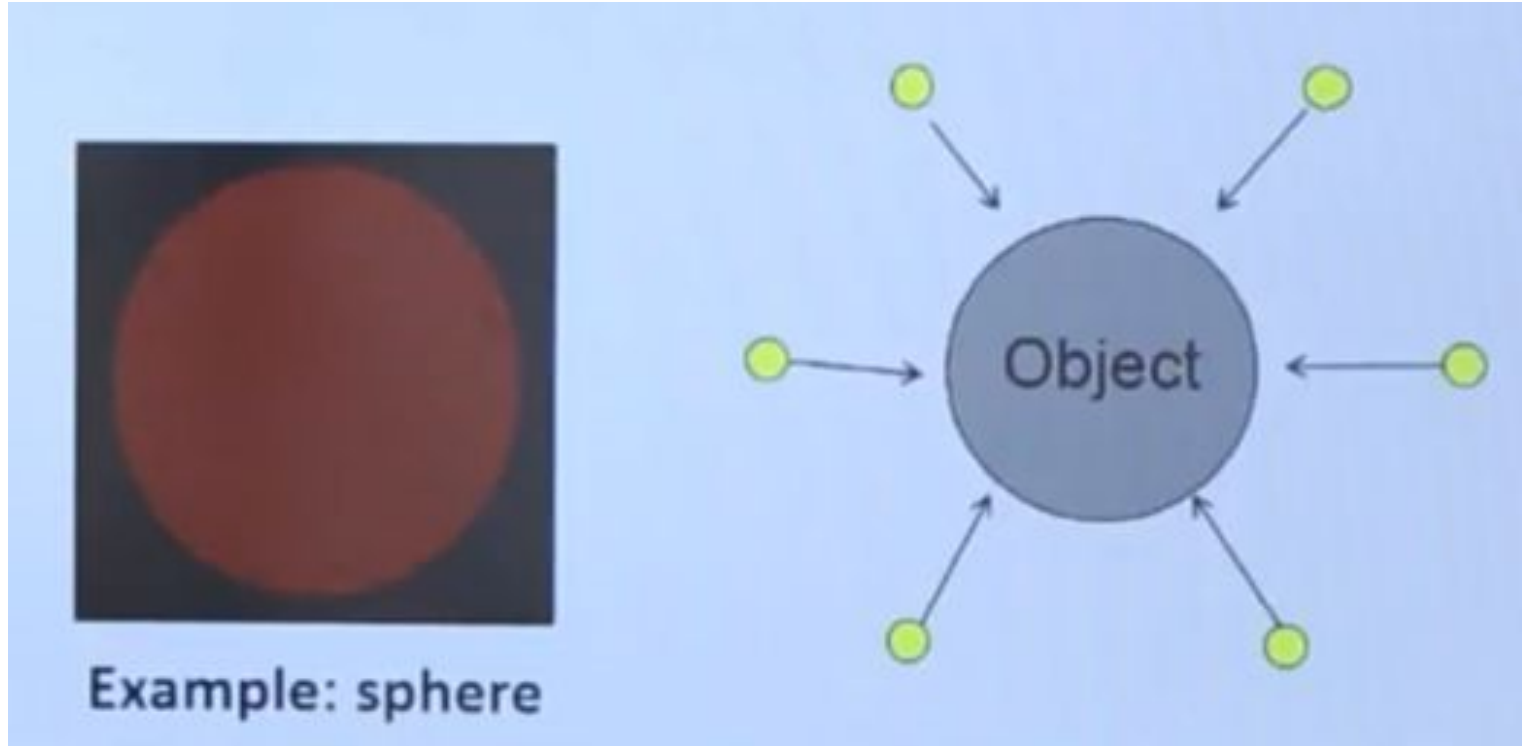
The reflected intensity I due to ambient light of any point on the surface is:

$$I = I_a K_a$$

Where, I_a = intensity of ambient light.

K_a = ambient reflection coefficient, $0 \leq K_a \leq 1$.

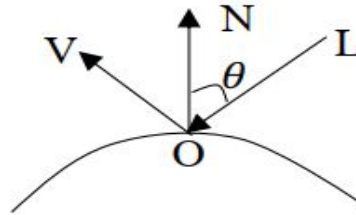
► Illumination Model ► **Ambient light** :



► Illumination Model ► Diffuse Reflection :

Diffuse reflection:

Light reflected with unequal intensity in all direction. Amount of light seen by viewer is independent of viewer direction. Brightness depends only on the angle θ between light direction and the surface normal.



L=light source

N=normal to surface

V=viewer direction

θ =angle betⁿ S & N

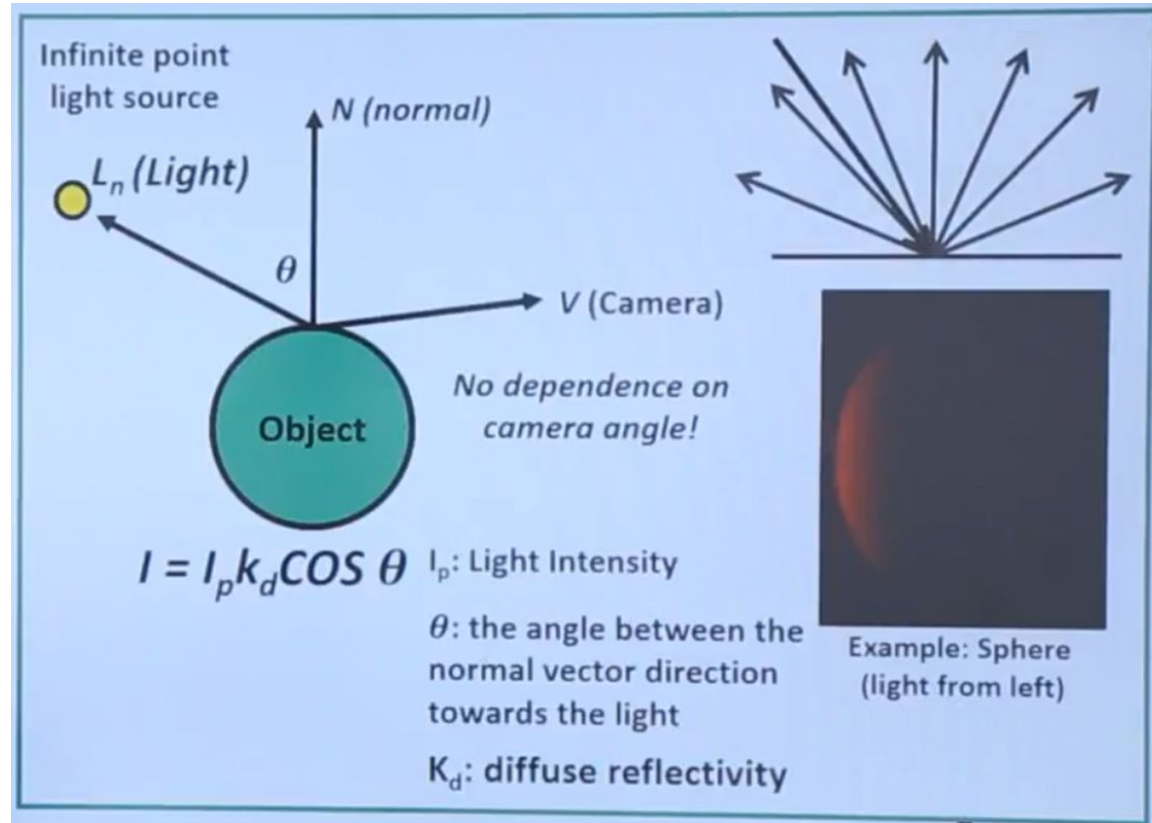
The reflected intensity I of any point on surface is

$$I = I_l K_d \vec{N} \cdot \vec{L} = I_l K_d \cos \theta$$

Where, K_d = coefficient of diffuse reflection

Net diffuse reflection = Diffuse reflection due to ambient light + Diffuse reflection due to light source
 $= I_{amb, diff} + I_{l, diff} = I_a K_a + I_l K_d \cos \theta$

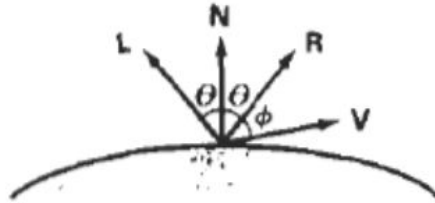
► Illumination Model ► Diffuse Reflection :



► Illumination Model ► **Specular reflection and Phong model** :

Specular reflection and Phong model

In shiny surface we see highlight or bright spot from certain viewing directions called specular reflection.



L=incident light

V=viewer

N=normal to surface

ϕ =angle betⁿ R & V

R=direction of ideal specular reflection

$$I_{l,spec} = K_s I_l (\vec{V} \cdot \vec{R})^{n_s}$$

Where, K_s = specular reflection coefficient

n_s = specular reflection parameter

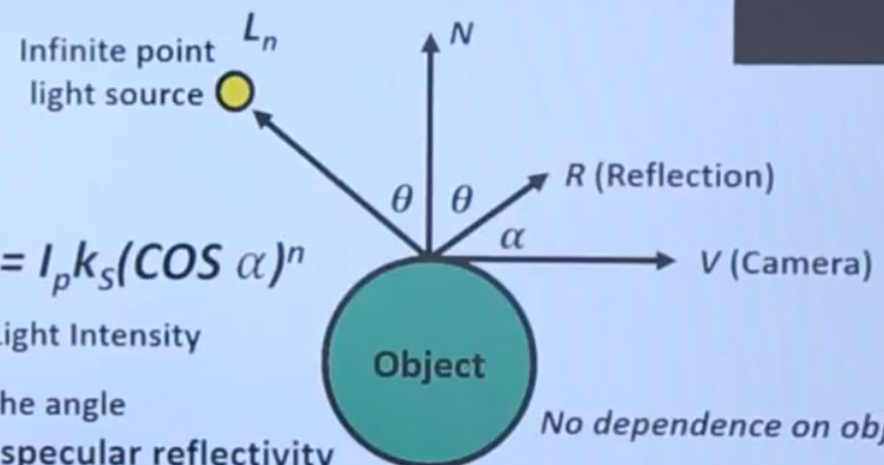
Phong model:

$$I_{l,spec} = K_s I_l \cos^{n_s} \phi$$

► Illumination Model ► **Specular reflection and Phong model** :

► Direct reflection of light source off shiny object

- Specular intensity n = shiny reflectance of object
- Result : specular highlight on object



$$I = I_p k_s (\cos \alpha)^n$$

I_p : Light Intensity
 α : the angle
 K_s : specular reflectivity

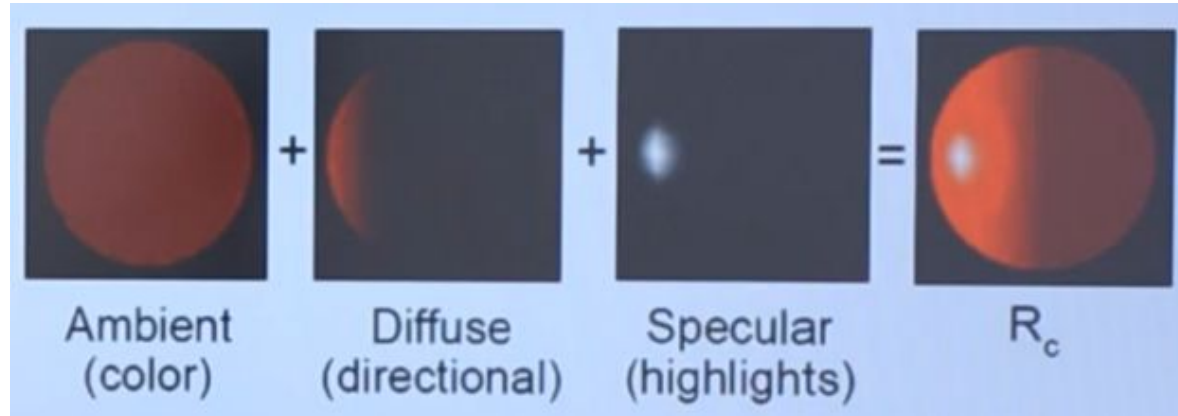
No dependence on object color.

► Illumination Model:

Combined diffuse and specular reflection

$$I = I_{diff} + I_{spec}$$

$$I = I_a K_a + I_l K_d (\vec{N} \cdot \vec{L}) + K_s I_l (\vec{V} \cdot \vec{R})^{n_s}$$



► Combining Lightening Models:

- Use dot product of the vector instead of calculating the angles θ, α

$$I_{\lambda} = I_a k_a + \sum_{p=1}^{lights} I_p [k_d (\vec{N} \cdot \vec{L}) k_s (\vec{V} \cdot \vec{R})^n]$$

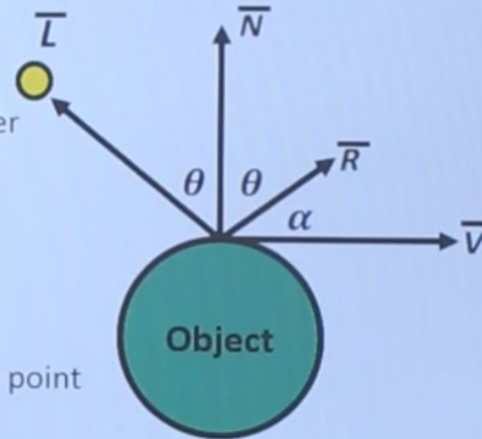
$\cos \theta$ $\cos \alpha$

$\vec{V}$: Vector from the surface to the viewer

$\vec{N}$: Normal vector at the colored point

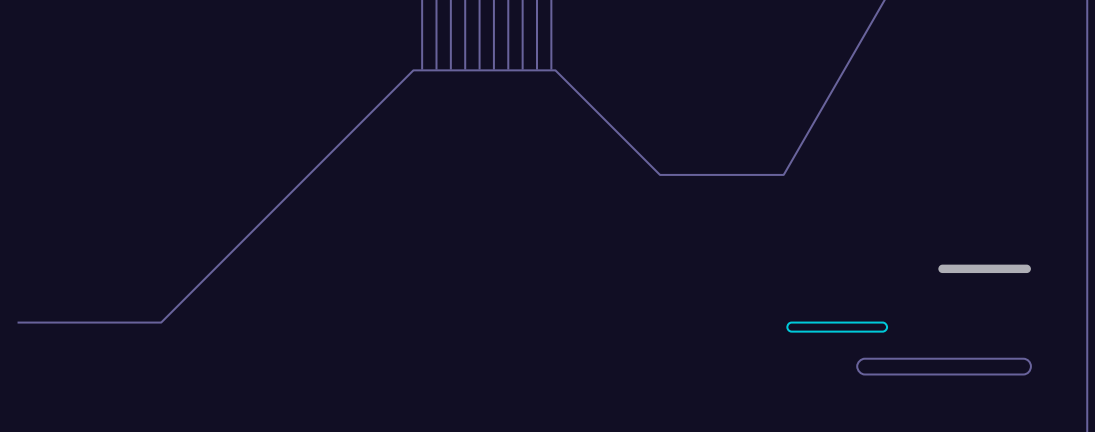
$\vec{R}$: Normalized reflection vector

$\vec{L}$: Normalized vector from the colored point towards the light source





Intensity Attenuation & Color Consideration, Transparency, Shadows



8.2

► Intensity Attenuation:

- ❖ Intensity is **attenuated** by the factor $1/d^2$, where d is the distance that the light has traveled.
- ❖ This means that a surface **close to the light source (small d)** receives a higher incident intensity from the source **than a distant surface** (large d).
- ❖ Graphics packages have a general inverse quadratic attenuation function

$$f(d) = \min\left(1, \frac{1}{a_0 + a_1d + a_2d^2}\right)$$

- ❖ User can fiddle with the coefficients a_0, a_1, a_2 to obtain a variety of lighting effects for a scene. The a_0 can be adjusted to prevent $f(d)$ from **becoming too large** when d is very small.
- ❖ This is an effective method for **limiting intensity values** when a **single light source** is used to illuminate a scene.

► Intensity Attenuation:

- ❖ With a given set of **attenuation coefficients**, we can limit the **magnitude of the attenuation function to 1** with the calculation

$$f(d) = \frac{1}{a_0 + a_1d + a_2d^2}$$

- ❖ Using this function, we can then write our basic illumination model as

$$I = K_a I_a + \sum_{i=1}^n f(d_i) I_{Li} [k_d (N \cdot L_i) + k_s (N \cdot H_i)^{n_s}]$$

Where d_i is the distance light has traveled from light source i

► Color Consideration:

- ❖ To incorporate color, we need to write the **intensity equation** as a function of the color properties of the light sources and object surfaces. Diffuse reflection coefficient vector for RGB component (k_{dR}, k_{dG}, k_{dB}) .
- ❖ For blue reflectivity component $k_{dR} = k_{dG} = 0$.

$$I_B = K_{aB}I_{aB} + \sum_{i=1}^n f(d_i)I_{LBi}[k_{dB}(N.L_i) + k_{sB}(N.H_i)^{n_s}]$$

- ❖ Surfaces are typically **illuminated with white light sources**, and in general we can set surface color so that the reflected light has **nonzero values** for **all three RGB components**. Calculated intensity levels for each color component can be used to **adjust the corresponding electron gun** in an RGB monitor.

Color specification with its spectral wavelength λ .

$$I_\lambda = K_a S_{a\lambda} I_{a\lambda} + \sum_{i=1}^n f(d_i) I_{L\lambda i} [k_d S_{d\lambda} (N.L_i) + k_s S_{s\lambda} (N.H_i)^{n_s}]$$

► Color Consideration:


$$I_{\lambda}^R = I_a^R k_a^R + \sum_{p=1}^{lights} I_p^R (k_d^R (\bar{N} \cdot \bar{L}) + k_s^R (\bar{V} \cdot \bar{R})^n)$$

$$I_{\lambda}^G = I_a^G k_a^G + \sum_{p=1}^{lights} I_p^G (k_d^G (\bar{N} \cdot \bar{L}) + k_s^G (\bar{V} \cdot \bar{R})^n)$$

$$I_{\lambda}^B = I_a^B k_a^B + \sum_{p=1}^{lights} I_p^B (k_d^B (\bar{N} \cdot \bar{L}) + k_s^B (\bar{V} \cdot \bar{R})^n)$$

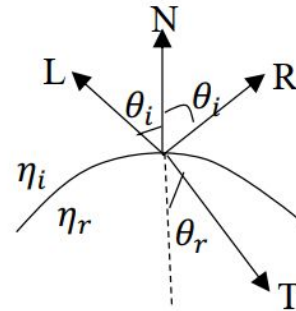
The RGB intensities thus calculated can be used to adjust the electron beam intensity of the three electron guns in any RGB monitor to generate a pixel in true color.

► Transparency:

- ❖ A transparent surface in general produces both reflected and transmitted light. The relative contribution of the transmitted light **depends on the degree of transparency** of the surface and whether any light sources are behind the transparent surface. **Both diffuse and specular transmission** can take place at the surfaces of a transparent object.

According to Snell's Law

$$\sin \theta_r = \frac{\eta_i}{\eta_r} \sin \theta_i$$



θ_i =angle of incidence

η_i =index of incidence

θ_r =angle of refraction

η_r =index of refraction

Fig: Reflection direction R and refraction direction T for a ray of light incident **upon** a surface with index of refraction η_r .

► Transparency:

Total surface intensity is calculated as

$$I = (1 - k_t)I_{reflect} + k_tI_{trans}$$

Where,

$I_{reflect}$ = reflected intensity

I_{trans} = transmitted intensity

k_t = transparency coefficient

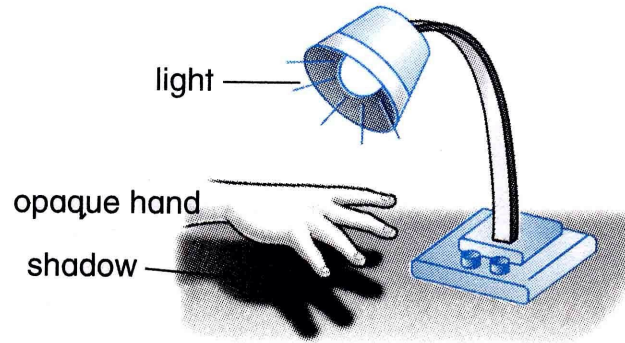
$(1 - k_t)$ = opacity factor

For highly transparent objects k_t is near 1.

For opaque objects k_t is near 0.

► Shadows:

- ❖ Hidden surface method can be used to **locate area where light sources** produce shadows. By applying a hidden surface method with a light source at the view position, we can determine **which surface sections** cannot be seen from the light source.
- ❖ Once we have determined the shadow area for all light sources, the shadows could be treated as **surface patterns** and **store in** pattern arrays.
- ❖ Shadow patterns generated by a hidden surface method **are valid** for any selected viewing position, as long as the light source positions **are not changed**.



8.3

► Polygon Rendering Method:

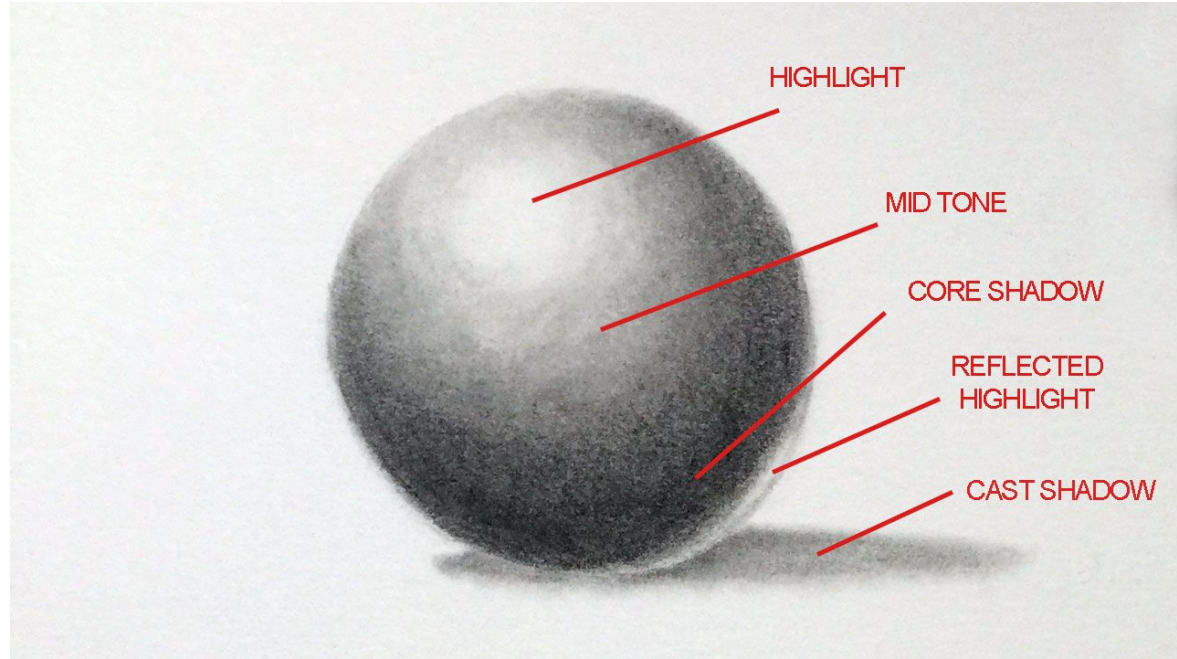
- ❖ Polygon Rendering is the process of calculating **intensity** and **color considerations** for a polygon.

Scene Description + Illumination Model + Rendering Technique = Image

- ❖ There are **2 ways** of polygon rendering:
 - Render each polygon surface with single intensity.
 - Calculate intensity at each point of the surface using interpolation scheme.
- ❖ In Order to produce a **realistic image** with various kinds of reflection, there are 3 kinds of **shading model** which are mainly applied to polygons:
 1. Constant intensity shading method
 2. Gouraud shading method (Intensity Interpolation)
 3. Phong shading method (Normal vector interpolation)

► Shading Method:

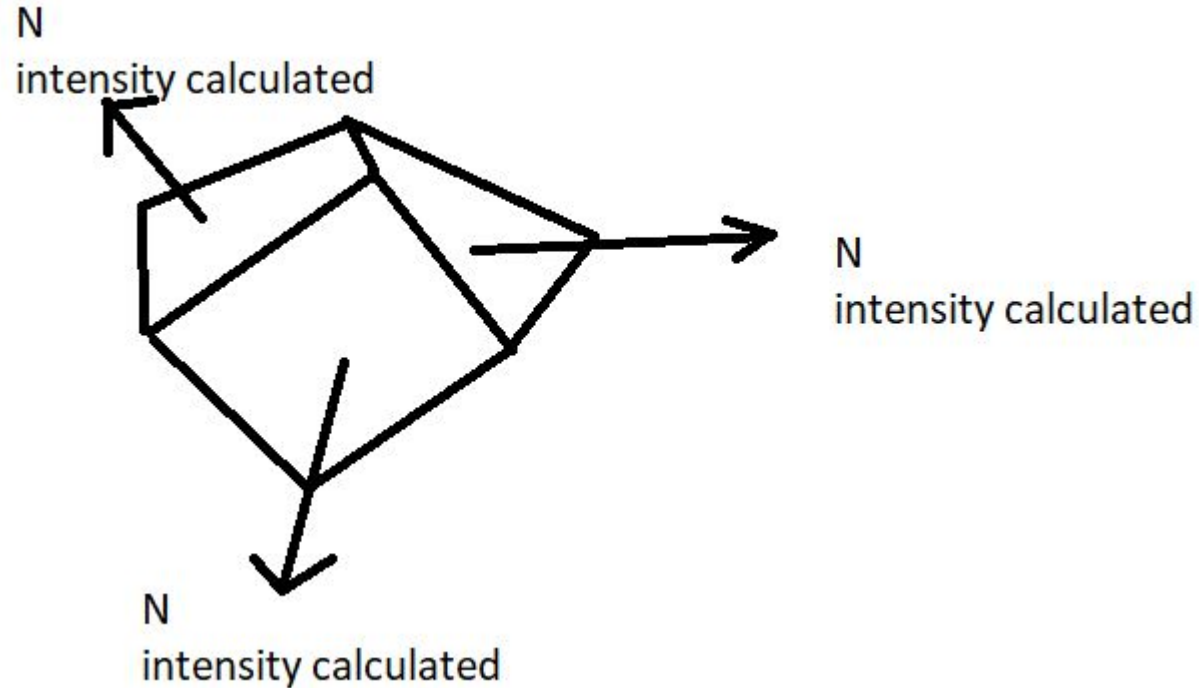
Shading is used traditionally in drawing for depicting a range of darkness by applying media more densely or with a darker shade for darker areas, and less densely or with a lighter shade for lighter areas.



► Flat / Constant Intensity Shading Method:

- ❖ A **fast and simple method** for rendering an object with polygon surfaces is constant intensity shading.
- ❖ In this method, a **single intensity** is calculated for **each polygon** and the intensity is applied to all the points of surface of polygon. Hence, all the points over the surface of the polygon are displayed with **same intensity value**.
- ❖ Constant shading is **useful for quickly** displaying the general appearance of a curved surfaces. This approach is **valid if**:
 - a. **Light source** is at infinity ($\vec{N} \cdot \vec{L}$ is constant on polygon).
 - b. **Viewer** is at infinity ($\vec{V} \cdot \vec{R}$ is constant on polygon).
 - c. **Object** is polyhedron and is not an approximation of an object with a curved surface.

► Flat / Constant Intensity Shading Method:



► Gouraud Shading Method (Intensity Interpolation):

- ❖ Developed by **Henri Gouraud**.
- ❖ It renders the polygon surface by **linearly interpolating intensity values** across the surface.
- ❖ Intensity values for each polygon are **matched** with the **values of adjacent polygon** along the **common edge**, thus eliminating the intensity discontinuities that occur in flat shading.
- ❖ **Algorithm:**

Each polygon surface is rendered with Gouraud Shading by performing following calculations:

1. Determine the **average unit normal vector** at each polygon vertex.
2. Apply an **illumination model** to each vertex to calculate the vertex intensity.
3. Linearly **interpolate the vertex** intensities over the surface of the polygon

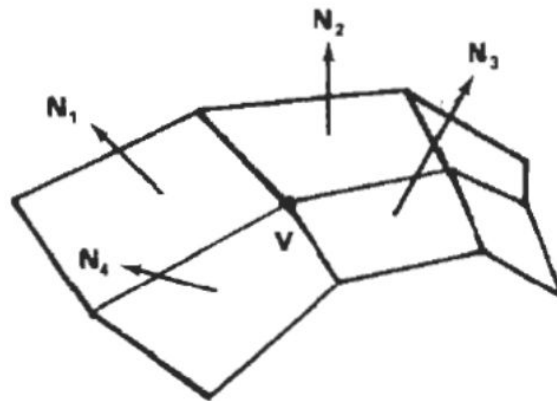
► Gouraud Shading Method (Intensity Interpolation):

Step 1: At each polygon vertex, we obtain a normal vertex by averaging the surface normals of all polygons sharing the vertex as:

$$N_v = \frac{\sum_{k=1}^n N_k}{|\sum_{k=1}^n N_k|}$$

Here in example: $N_v = \frac{N_1 + N_2 + N_3 + N_4}{|N_1 + N_2 + N_3 + N_4|}$

Where N_v is normal vector at a vertex sharing 4 surfaces as in figure.

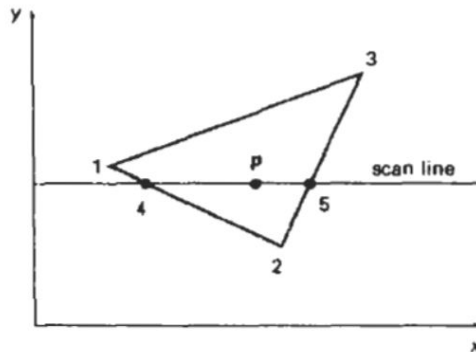


► Gouraud Shading Method (Intensity Interpolation):

Step-2: Apply an **illumination model** to each vertex to calculate the **vertex intensity**.

Step-3: **Linearly interpolate** the vertex intensities over the surface of the polygon

Once we know N_v , intensity at the vertices can obtain from lightning model.



- ❖ In figure, the intensity of vertices 1, 2, 3 are I_1, I_2, I_3 , which are obtained by averaging normals of each surface sharing the vertices and applying a illumination model. For each scan line, intensity at intersection of line with Polygon edge is linearly interpolated from the intensities at the edge end point.

► Gouraud Shading Method (Intensity Interpolation):

So intensity at point 4 is to interpolate between intensities I_1 and I_2 using only the vertical displacement of the scan line:

$$I_4 = \frac{y_4 - y_2}{y_1 - y_2} I_1 + \frac{y_1 - y_4}{y_1 - y_2} I_2$$

Similarly, the intensity at point 5 is obtained by linearly interpolating intensities at I_2 and I_3 as

$$I_5 = \frac{y_5 - y_2}{y_3 - y_2} I_3 + \frac{y_3 - y_5}{y_3 - y_2} I_2$$

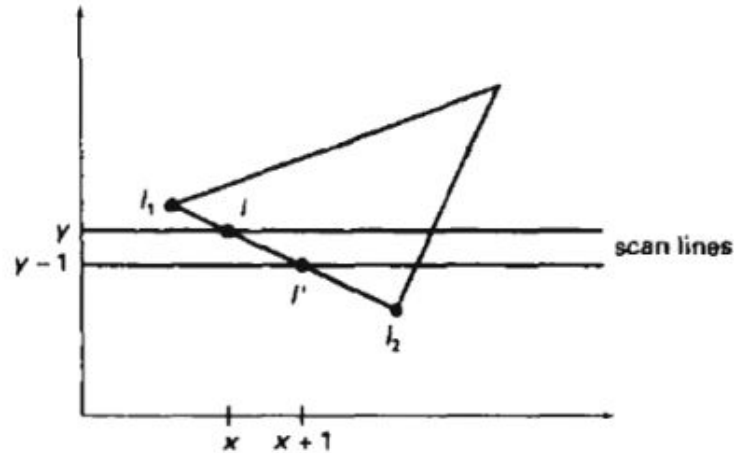
The intensity of a point P in the polygon surface along scan-line is obtained by linearly interpolating intensities at I_4 and I_5 as,

$$I_p = \frac{x_5 - x_p}{x_5 - x_4} I_4 + \frac{x_p - x_4}{x_5 - x_4} I_5$$

Then incremental calculations are used to obtain Successive edge intensity values between scan-lines as and to obtain successive intensities along a scan line. As shown in Fig. below, if the intensity at edge position (x, y) is interpolated as:

$$I = \frac{y - y_2}{y_1 - y_2} I_1 + \frac{y_1 - y}{y_1 - y_2} I_2$$

► Gouraud Shading Method (Intensity Interpolation):



Then, we can obtain the intensity along this edge for next scan line at $y-1$ position as

$$I' = \frac{y-1-y_2}{y_1-y_2} I_1 + \frac{y_1-(y-1)}{y_1-y_2} I_2 = I + \frac{I_2 - I_1}{y_1 - y_2}$$

Similar calculations are made to obtain intensity successive horizontal pixel.

► Gouraud Shading Method (Intensity Interpolation):

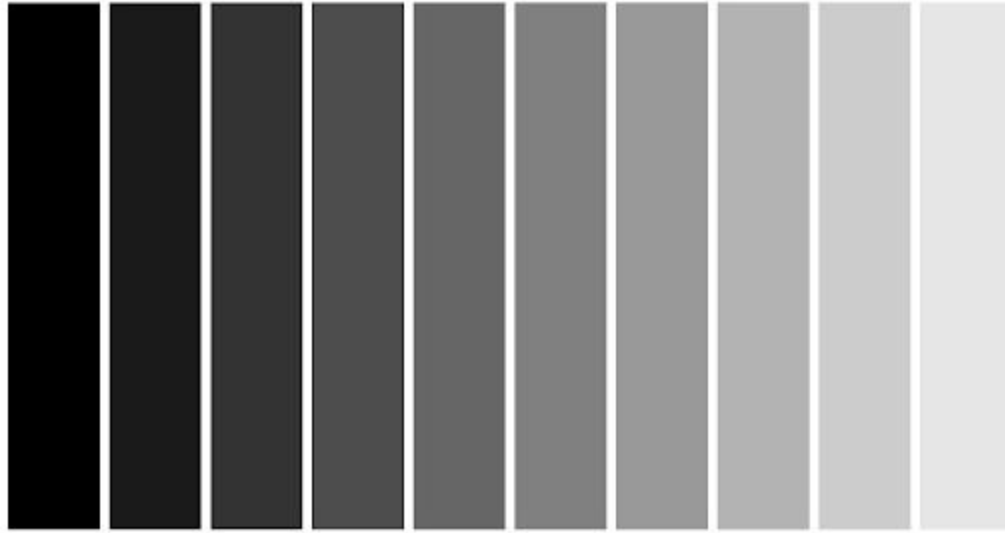
❖ Advantages:

- It removes the intensity discontinuity which exists in constant shading model.
- It can be combined with hidden surface algorithm to fill in the visible polygon along each scan line.

❖ Disadvantages:

- Causes bright or dark intensity streak called mach bands.

► Match Band Effect:



► Phong Shading Method (Normal Vector Interpolation):

- ❖ Developed by **Phong Bui Tuong**.
- ❖ It renders the polygon surface by **linearly interpolating normal vector** across the surface.
- ❖ **A more accurate method** for rendering a polygon surface is to **interpolate normal vector** and then **apply illumination model** to each surface point.
- ❖ It displays more realistic highlights and greatly **reduces the mach-band effect**.
- ❖ **Algorithm:**

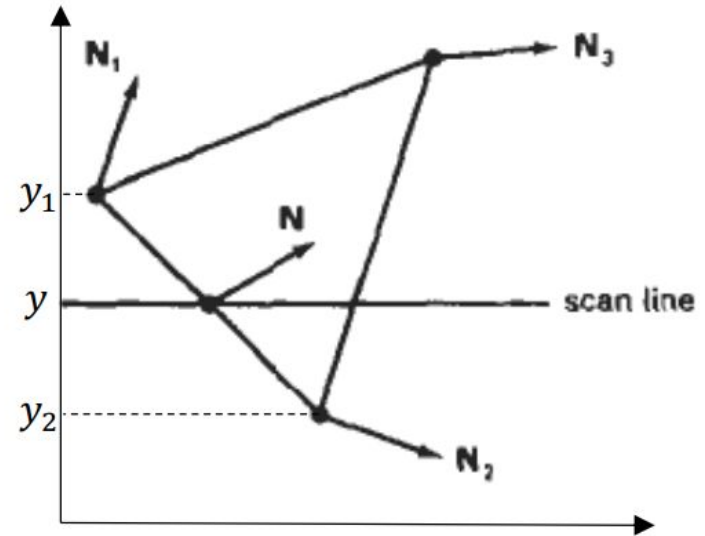
A polygon surface is rendered with Phong shading by carrying out following calculations.

- Determine the **average normal unit vectors** at each polygon vertex.
- **Linearly interpolate** vertex normals over the surface of polygon.
- **Apply illumination model** along each scan line to calculate projected pixel intensities for the surface points.

► Phong Shading Method (Normal Vector Interpolation):

In figure, N_1 , N_2 , N_3 are the normal unit vectors at each vertex of polygon surface. For scan-line that intersect an edge, the normal vector N can be obtained by vertically interpolating normal vectors of the vertex on that edge as.

$$N = \frac{y - y_2}{y_1 - y_2} N_1 + \frac{y_1 - y}{y_1 - y_2} N_2$$



Incremental calculations are used to evaluate normals between scan lines and along each individual scan line as in Gouraud shading. Phong shading produces accurate results than the direct interpolation but it requires considerably more calculations.

► Phong Shading Method (Normal Vector Interpolation):

❖ Advantages:

- It displays more realistic highlights on a surface.
- It greatly reduces the Mach band effect.
- It gives more accurate results.

❖ Disadvantages:

- It requires more calculation and greatly increases the cost of shading steeply.

► Shading Final Output- Comparison:

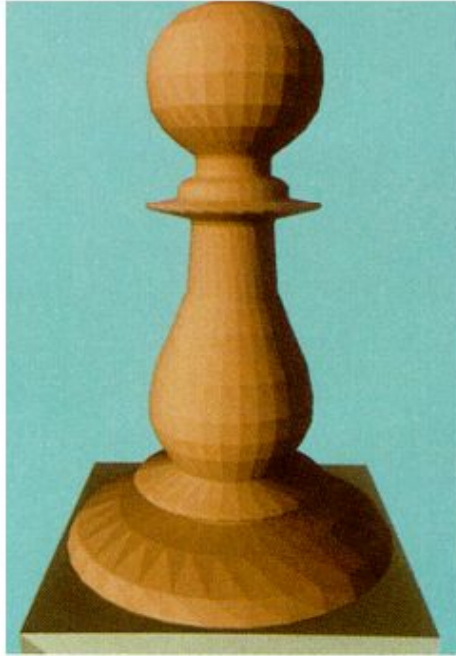


Figure 1: Flat shading.

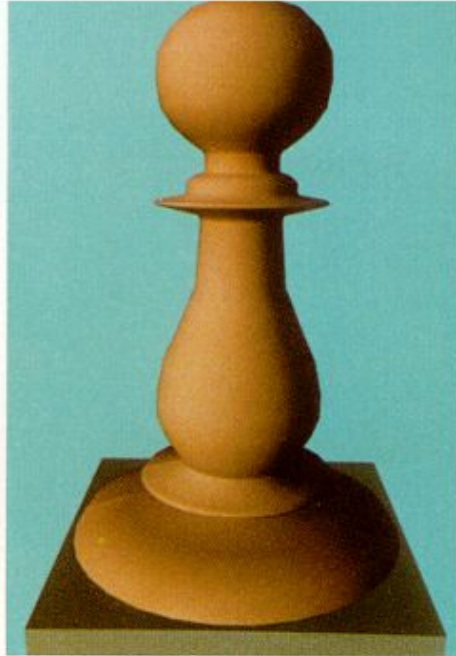


Figure 2: Gouraud shading.

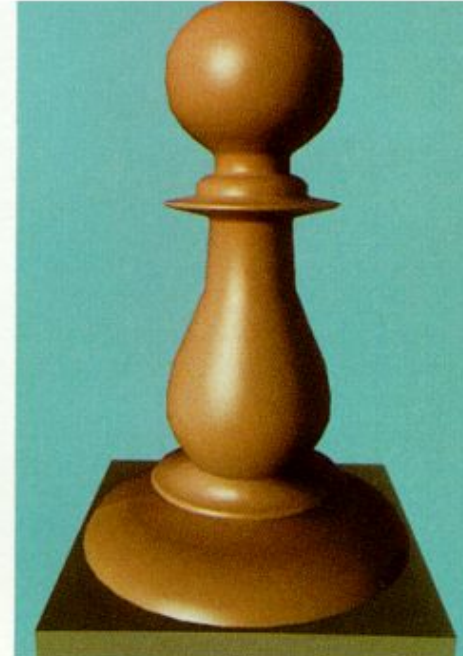


Figure 3: Phong shading.

► Fast Phong Shading :

- ❖ Fast phong shading approximates the intensity calculations using a **Taylor-series expansion and triangular surface patches**.
- ❖ **Surface normal** at any point (x, y) over a triangle as; $N = Ax + By + C$

A, B, C are determined from the three vertex equations:

$$N_k = Ax_k + By_k + C, \quad k = 1, 2, 3$$

With (x_k, y_k) denoting a vertex position.

Omitting the reflectivity and attenuation parameters,

$$I_{diff}(x, y) = \frac{L \cdot N}{|L||N|} = \frac{L \cdot (Ax + By + C)}{|L||Ax + By + C|} = \frac{(L \cdot A)x + (L \cdot B)y + L \cdot C}{|L||Ax + By + C|}$$

► Fast Phong Shading :

We can write this,

$$I_{diff}(x, y) = \frac{ax + by + c}{(dx^2 + exy + fy^2 + gx + hy + i)^{1/2}} \dots \dots \dots (i)$$

Where, a, b, c, d are used to represent the various dot product. For e.g.

$$a = \frac{L \cdot A}{|L|}$$

Finally, denominator in eq. (i) can be expressed as a Taylor-series expansion and retain terms up to second degree in x & y. This yields

$$I_{diff}(x, y) = T_5x^2 + T_4xy + T_3y^2 + T_2x + T_1y + T_0$$

Where, each T_k is a function of parameters a, b, c, and so forth.

This method still takes twice as long as in Gouraud shading. Normal Phong shading takes six to seven times that of Gouraud shading.

► Fast Phong Shading :

Performing the indicated dot products and expanding the vector magnitude yields

$$I_{diffuse}(x, y) = \frac{ax + by + c}{\sqrt{dx^2 + exy + fy^2 + gx + hy + i}}$$

with

$$a = \frac{\vec{L} \cdot \vec{A}}{|\vec{L}|},$$

$$b = \frac{\vec{L} \cdot \vec{B}}{|\vec{L}|},$$

$$c = \frac{\vec{L} \cdot \vec{C}}{|\vec{L}|},$$

$$d = \vec{A} \cdot \vec{A},$$

$$e = 2\vec{A} \cdot \vec{B},$$

$$f = \vec{B} \cdot \vec{B},$$

$$g = 2\vec{A} \cdot \vec{C},$$

$$h = 2\vec{B} \cdot \vec{C}, \text{ and}$$

$$i = \vec{C} \cdot \vec{C}.$$

► Flat vs Phong vs Gouraud Shading :

Flat shading	Gouraud shading	Phong shading
Computes illumination once per polygon and apply it to whole polygon.	Computes illumination at vertices and interpolates.	Applies illumination at every point of polygon surface.
Creates discontinuous in color.	Interpolates colors along edges and scan line.	Interpolates normal instead of colors.
Problems of Mach bands.	Handles Mach bands problem found in flat shading.	Removes Mach bands completely.
Low cost.	Not so expensive.	More expensive than gouraud shading.
Requires very less processing and is fast in time.	Required moderate processing time.	Requires complex processing and is slower but is more efficient as compared to other shading methods.
Lighting equation used once per polygon.	Lighting equation used at each vertex.	Lighting equation used at each pixel.

► Assignment-8:

Deadline: **27th February, 2026**

1. How a realistic image can be generated in computer graphics? Explain fast phong shading. **2080**
2. Define the term 'rendering' in computer graphics. Explain Phong Shading Method with its advantage and disadvantage. **2079**
3. What is the advantage of real time rendering over offline rendering? **2078**
4. Define intensity attenuation. Distinguish between Gouraud shading and Phong shading model. **2078**
5. What is specular reflection model of phong? Explain shadows and intensity attenuation in illumination modelling?
6. Define realism in human perception. What is the significance difference between rendering and image synthesis in creating computer generated 3D image? Describe any two polygon rendering methods. **2077**
7. How can we detect shadows in computer graphics? List the challenged in computing light model.
8. What do you mean by shade? Explain Gouraud shading method in detail with proper illustration.
9. What do you mean by illumination & intensity. Explain different illumination models.



▶ **THANKS!**

Do you have any questions?

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